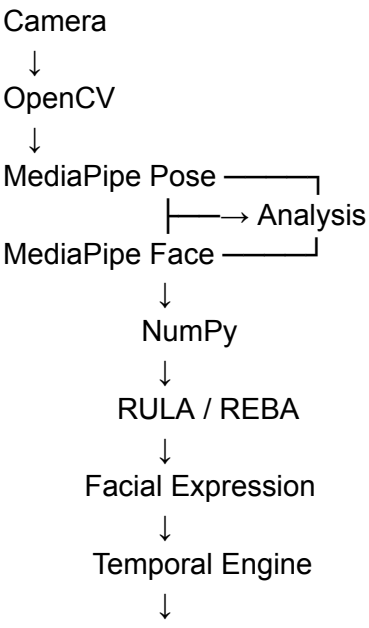
For the **reworked EdgeErgo PRD**, the tech stack is intentionally kept lightweight and divided into **Edge Agent** and **Central Server**.

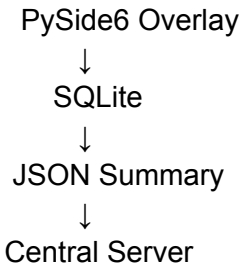
1. Edge Agent — Worker PC

This is installed on **every worker's computer**.

Technology	Purpose
Python	Main Edge Agent language
OpenCV	Camera access, frame capture, resizing, preprocessing
MediaPipe Pose	Human pose/body landmark detection
MediaPipe Face	Face landmarks/features and facial-expression analysis
NumPy	Angle calculation, vectors, numerical processing
Python RULA/REBA rules	Ergonomic scoring and risk classification
SQLite	Local storage and offline synchronization queue
PySide6	Worker desktop overlay and notifications

Edge flow





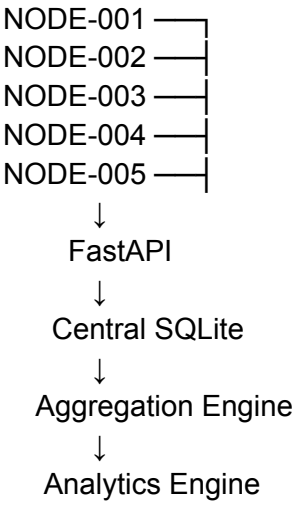
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2. Central Server

There is **one central server/PC** for collecting data from all Edge Nodes.

Technology	Purpose
Python	Central backend language
FastAPI	REST API and Edge-node communication
SQLite	Central database for the initial implementation
Python Aggregation Engine	Combines data from all workstations
Python Analytics Engine	Collective trends and statistics
Python CSV generation	Export collected data
Python Report Generator	Detailed reports

Architecture:



↓
Reports / CSV

---

3. Central Dashboard

The administrator dashboard uses:

Technology	Purpose
React	Dashboard UI
Vite	React development/build tooling
TypeScript	Frontend programming language
Tailwind CSS	UI styling
Chart library	Risk/trend visualization

React is **only for the Central Dashboard**.

You do **not** need React on every worker computer.

---

4. Communication

For communication between Edge PCs and the Central Server:

HTTP/HTTPS
+
REST API
+
JSON

Example:

Worker PC
│
│ POST /api/v1/edge/assessment
│

▼
FastAPI Server

The transmitted information is something like:

```
{
  "device_id": "NODE-004",
  "avg_rula": 4.2,
  "max_rula": 6,
  "avg_reba": 5.1,
  "risk_level": "MODERATE",
  "dominant_posture": "NECK_FLEXION",
  "expression": "NEUTRAL"
}
```

Not:

❌ Video
❌ Camera frames
❌ Face images
❌ Screenshots

---

5. Deployment / Production Stack

For making it testable on other computers:

Worker

Python Application
↓
PyInstaller
↓
EdgeErgoAgent.exe

So a worker doesn't need to manually install Python, OpenCV, MediaPipe, etc.

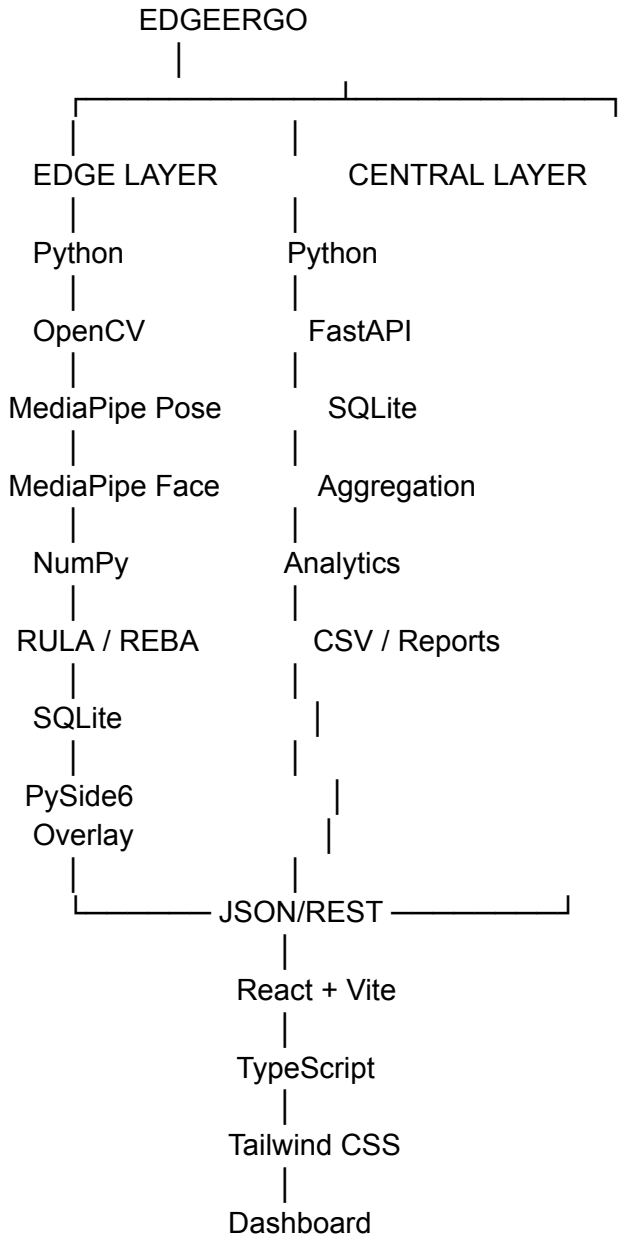
Central Server

FastAPI
+
Built React frontend
+
SQLite

The administrator can access the dashboard through a browser.

---

6. Complete Tech Stack



In one line

Edge: `Python + OpenCV + MediaPipe Pose + MediaPipe Face + NumPy + RULA/REBA + SQLite + PySide6`

Server: `Python + FastAPI + SQLite + Aggregation/Analytics + CSV/Report Generator`

Dashboard: `React + Vite + TypeScript + Tailwind CSS + Charts`

Communication: `REST API + JSON + HTTP/HTTPS`

Deployment: `PyInstaller (Edge Agent) + Production React build + FastAPI`

This is the stack I would keep for EdgeErgo because it avoids unnecessarily heavy technologies while still giving you a genuinely distributed Edge AI architecture.